[20] Non-Convex Optimization for Machine Learning

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Non-convex Optimization for Machine Learning — Background for RQ1, RQ2, RQ3

Problem: How can nonconvex algorithms have rigorous guarantees when arbitrary nonconvex functions admit poor minima and difficult saddles? **Approach:** A *monograph* organizing tractable machine-learning structures—sparsity, low rank, projections and strict-saddle geometry—and their optimization algorithms. **Main message:** Strong results come from *structural assumptions on a problem class*, not from nonconvexity itself being harmless.

1. A nonconvex constraint is not the same as a bad objective

Consider least squares with sparsity:

$$\min_{w\in\mathbb{R}^d} \frac{1}{2}\|Aw-b\|^2 \qquad \text{subject to} \qquad \|w\|_0 \le s.$$

The quadratic loss is convex in w, but the feasible set of vectors with at most s nonzero entries is nonconvex. Taking the average of two sparse vectors can yield more than s nonzeros. The optimization challenge here lies partly in the *constraint geometry*.

2. Projected gradient descent from first principles

A projected step takes a gradient descent move and maps it back into the feasible set $\mathcal{C}$:

$$\boxed{w_{t+1} = \Pi_{\mathcal{C}}(w_t - \eta\nabla f(w_t))}, \qquad \Pi_{\mathcal{C}}(z) \in \arg\min_{u\in\mathcal{C}} \|u-z\|.$$

For the sparse set $\mathcal{C} = \{w : \|w\|_0 \le s\}$, Euclidean projection keeps the s coordinates of largest magnitude and zeros the rest. For a rank-r matrix constraint, projection keeps the top r singular components (a truncated SVD). These are exact, concrete examples in which nonconvex projections are computationally manageable.

3. Why restricted curvature appears

An unrestricted loss may be badly conditioned in some directions. Yet the algorithm only moves along differences of feasible points. A useful restricted-geometry condition for low-complexity directions h has the form

$$\alpha\|h\|^2 \le h^\top\nabla^2 f(w)h \le \beta\|h\|^2 \quad \text{for relevant structured } h.$$

This is a *schematic reminder* of restricted strong convexity/smoothness: the set of relevant directions and constants depends on the particular sparse/low-rank problem. It explains why convergence can be proved without global strong convexity.

4. Saddles and the role of curvature

A smooth critical point has $\nabla f(w) = 0$. If the Hessian has a negative eigenvalue, $\lambda_{\min}(\nabla^2 f(w)) < 0$, a second-order Taylor expansion implies a direction of local decrease. The monograph reviews strict-saddle conditions and algorithms that inject randomness or exploit curvature to avoid undesirable stationary points. For general deep ReLU networks, neither such geometric structure nor smoothness is automatic.

Why this matters for RQ4. Matrix sensing or sparse recovery can admit strong global-convergence statements because their data operators satisfy restricted isometry/incoherence-like properties. We should not transport those guarantees directly to arbitrary neural-network losses.

5. Scope and limitations

This is an instructional synthesis of several research lines, not a single original theorem. Its positive convergence results depend on the specific problem, feasible set, sampling conditions, initialization and regularity assumptions. It is valuable for interpreting how *special geometry* rescues nonconvex optimization, while reminding us that broad guarantees for realistic neural nets remain substantially harder. **Our takeaway.** The main lesson is to ask "which structural property makes this nonconvex problem tractable?" before claiming that gradient descent solves nonconvex problems generally. **Source:** https://www.cse.iitk.ac.in/users/purushot/papers/ncom-book.pdf (Parts I–III, especially Chapters 3 and 6).